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Reg. No. | | | | | | | | | | |

[22BS1MA01]

FIRST SEMESTER B.Tech EXAMINATION, FEBRUARY - 2023

CALCULUS AND MATRIX ALGEBRA

Time: 3 Hours **Maximum Marks: 100**

Instructions:

i. Answer all questions.

ii. Missing data may be suitably assumed.

PART - A **5 X 2=10**

1.	a.	Evaluate the integral $\int_0^{\pi} sin^4 x\, dx$.	L2	CO1	02
	b.	If $u = \log(x^2 + y^2)$ then find $\frac{\partial^2 u}{\partial x \partial y}$.	L2	CO1	02
	c.	A particle moves along a curve whose parametric equations are: $x = e^{-t}$, $y = 2\cos 3t$, $z = 2\sin 3t$, where t is the time variable. Find acceleration at any time t and also its magnitude at $t = 0$.	L2	CO1	02
	d.	Find the rank of matrix $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 1 & 3 & 4 \end{bmatrix}$.	L2	CO1	02
	e.	Solve the differential equation $(D^2 - 3D + 2)y = 0$.	L2	CO5	02
		PART - B		**6 X 5=30**	
2.	a.	Evaluate the integral $\int_0^{\pi} \frac{sin^4 x}{(1+cos\, x)^2} dx$.	L2	CO4	05
		OR			
	b.	Evaluate the integral $\int_0^{\pi/2} \sqrt{cot\, \theta}\, d\theta$.	L2	CO4	05
3.	a.	Prove that $\frac{\partial^2 z}{\partial t^2} = c^2 \frac{\partial^2 z}{\partial x^2}$, where $z = f(x + ct) + \varphi(x - ct)$.	L2	CO2	05
		OR			
	b.	Show that $\frac{\partial(u,v,w)}{\partial(x,y,z)} = 4$, where $u = \frac{yz}{x}$, $v = \frac{zx}{y}$ and $w = \frac{xy}{z}$.	L2	CO2	05
4.	a.	Find directional derivative of the function $f = x^2yz + 4xz^2$ at $(1, -2, -1)$ along the direction of the vector $2\hat{\imath} - \hat{\jmath} - 2\hat{k}$.	L2	CO6	05
		OR			
	b.	Find angle between the surfaces $x^2 + y^2 + z^2 = 9$ and $z = x^2 + y^2 - 3$ at the point $(2, -1, 2)$.	L3	CO6	05
5.	a.	Test for consistency and hence solve the system of equations: $x + y + z = 6$, $x - y + 2z = 5$, and $3x + y + z = 8$.	L2	CO3	05
		OR			
	b.	Solve the given system of equations by Gauss-elimination method: $2x + y + z = 10$, $3x + 2y + 3z = 18$ and $x + 4y + 9z = 16$.	L2	CO3	05
6.	a.	Solve the differential equation $\frac{d^3y}{dx^3} - 3\frac{d^2y}{dx^2} + 3\frac{dy}{dx} - y = 0$.	L3	CO5	05

		OR			
	b.	Solve the differential equation $y'' - 4y' + 13y = \cos 2x$.	L3	CO5	**05**
7.	**a.**	Solve the differential equation $y'' + 2y' + y = 2x + x^2$.	L3	CO5	**05**
		OR			
	b.	Explain MATLAB code for first and second order partial derivatives of the function $f(x,y) = e^x \cos y + e^y \sin x$.	L2	CO2	**05**
		PART - C		**6 X 10=60**	
8.	**a.**	Show that $\int_0^{\pi/2} \sqrt{\sin\theta}\, d\theta \times \int_0^{\pi/2} \frac{1}{\sqrt{\sin\theta}} d\theta = \pi$.	L3	CO4	**10**
		OR			
	b.	i. Discuss various commands of multiple assignments using MATLAB code and provide example for each of them. ii. Write MATLAB code to evaluate the integral $\int_0^{\pi} (\sin y - 5y^3) dy$	L2	CO4	**10**
9.	**a.**	Find extreme values of the function $f(x,y) = x^4 + y^4 - 2x^2 + 4xy - 2y^2$.	L3	CO6	**10**
		OR			
	b.	Temperature T at any point (x,y,z) in the space is given by $T = 400xyz^2$. Calculate the highest temperature at the surface of a unit sphere $x^2 + y^2 + z^2 = 1$.	L3	CO6	**10**
10.	**a.**	Find unit tangent vector and unit normal vector to the curve $\vec{r}(t) = \cos 2t\hat{i} + \sin 2t\hat{j} + t\hat{k}$ at $x = \frac{1}{\sqrt{2}}$.	L3	CO6	**10**
		OR			
	b.	Show that the vector point function $\vec{v} = \frac{x\hat{i}+y\hat{j}}{x^2+y^2}$ is both solenoidal and irrotational.	L3	CO6	**10**
11.	**a.**	Diagonalize the matrix $\begin{bmatrix} 11 & -4 & -7 \\ 7 & -2 & -5 \\ 10 & -4 & -6 \end{bmatrix}$.	L3	CO3	**10**
		OR			
	b.	Investigate the values of λ and μ so that the system of equations $x + y + z = 6$, $x + 2y + 3z = 10$ and $x + 2y + \lambda z = \mu$ will have **i.** no solution **ii.** a unique solution **iii.** infinite number of solutions	L3	CO3	**10**
12.	**a.**	Solve the system $X' = \begin{pmatrix} 4 & 2 \\ 2 & 1 \end{pmatrix} X + \begin{pmatrix} 3e^t \\ e^t \end{pmatrix}$ by diagonalization method and hence discuss its stability.	L3	CO5	**10**
		OR			
	b.	Solve the differential equation $(D^2 + 4D + 5)y = e^{-2x} \cos x$.	L3	CO5	**10**

13.	a.	i. Explain any five elementary MATLAB built-in mathematical functions. ii. Write MATLAB code for $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial^2 f}{\partial x^2}, \frac{\partial^2 f}{\partial y^2}$, given $f(x,y) = x^3 + 3xy - 15x^2 - 15y^2 + 72x - 4y.$	L2	CO2	10
		OR			
	b.	Explain MATLAB code to create the matrix $A = \begin{bmatrix} -1 & 0 & 4 \\ 2 & 6 & 1 \\ 0 & 2 & 3 \end{bmatrix}$ and also to perform following operations: i. Extract 3rd column elements ii. Determinant of A iii. Rank of A iv. Determine Eigen values and Eigen vectors.	L3	CO3	10